

Individual Assignment 1

Pilgrim Bank Cases Study (A): Nonlinear Machine Learning Models for Customer Profit Prediction at Pilgrim Bank

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Abstract

In this study, I revisited the Pilgrim Bank (A) case study 1, using nonlinear machine learning models to evaluate and predict customer profitability. Initially, our original analysis relied on linear regression, which assumes additive and linear relationships. In this expanded study, I aim to investigate whether modern nonlinear algorithms such as Decision Tree, Random Forest, Support Vector Mechanism (SVM), and XGBoost offer improved predictive accuracy and deeper insights. The analysis follows a structured process including comprehensive data preparation, feature engineering, hyperparameter tuning, cross-validation, model evaluation, and interpretability analysis. My results show that nonlinear models significantly outperform linear regression, in which XGBoost demonstrates the strongest performance in capturing complex nonlinear relationships and interactions among demographic, behavioral, and geographic variables. This expanded narrative provides detailed methodological reasoning, an interpretation of the results, and implications for managerial decision-making.

1. Introduction

The Pilgrim Bank case investigates whether customer adoption of online banking services is linked to changes in profitability. Traditionally, this analysis was approached using linear regression models, which assume simple linear relationships between predictors and profits. However, customer behavior in financial contexts often exhibits complexity, with nonlinear effects, threshold behaviors, interaction patterns, and high variability across demographic groups that can impact customer behavior. As banking decisions increasingly rely on granular customer analytics, machine learning provides a robust suite of nonlinear modeling tools that can capture richer patterns than classical statistical models.

1.1.Objectives

The purpose of this expanded report is to apply modern nonlinear regression algorithms to the same Pilgrim Bank dataset to determine whether these models can produce more accurate predictions of customer profitability. By comparing nonlinear models to the linear regression baseline, this study aims to reveal the advantages and trade-offs associated with using more flexible learning algorithms within a financial analytics context. Hence, the goals of this study are to compare performance, discuss findings, and reflect on whether these models improve on the original baseline.

2. Data Preparation and Feature Engineering

In the data analysis process, data preparation is a crucial step in any predictive modeling exercise. Especially when employing machine learning algorithms that are sensitive to missing values, variable scales, and inconsistent data formats. The Pilgrim Bank dataset contains detailed customer-level information, including profit, age, income, tenure, district, bill payment behavior, and online usage. Before modeling with machine learning predictive algorithms, I consider several steps to ensure that the data are analytically ready.

2.1. Mean imputation for numerical variables and feature-engineering

First, I examine the missing values. The dataset contained incomplete records for age and income, which can introduce bias if not handled appropriately. To preserve information, two complementary strategies are applied. Missing values are imputed using mean imputation for numeric fields, ensuring that all algorithms receive complete and consistent data. In addition, missingness indicator variables (`age_exists` and `income_exists`) are created to allow the models to detect potential patterns related to missing data itself. This

process is a valuable feature-engineering technique when the absence of a value that may be correlated with behavior or profitability is identified.

2.2. One-hot encoding for district categories and scaling for numerical fields

Categorical variables, such as districts, are transformed using one-hot encoding, producing binary indicator columns. Numeric variables, including age, income, tenure, and bill pay, are scaled using standardization, which is particularly important for algorithms such as SVM and XGBoost that are sensitive to the magnitude of feature values. These preprocessing steps are encapsulated in a Column Transformer pipeline to ensure reproducibility and consistency across all models.

2.3. 80/20 train-test split and Unified Column Transformer pipeline

In this step, I split the dataset into training and testing sets using an 80/20 split. The training set is used for model fitting and hyperparameter tuning through cross-validation, while the test set provides an unbiased assessment of the final model's performance. By combining imputation, scaling, encoding, and engineered indicators into a unified pipeline, the dataset is thoroughly prepared for robust nonlinear modeling.

3. Modeling Approach and Hyperparameter Tuning

In this study, I consider four nonlinear machine learning models based on their suitability for structured financial data and their ability to capture complex relationships. Each model is tuned using GridSearchCV with five-fold cross-validation to ensure robust generalization. The Decision Tree model serves as the simplest nonlinear baseline, capturing hierarchical splits and interactions; tuning is focused on depth and minimum sample constraints to control overfitting. Then, Random Forest, an ensemble of decision

trees, is included for its ability to reduce variance. Key hyperparameters, such as the number of trees and tree depth, are optimized to enhance stability and predictive accuracy. Support Vector Regression (SVR) or SVM with an RBF kernel is then incorporated to model nonlinear boundaries, enabling the detection of curved and irregular relationships between features and profitability. XGBoost, a high-performance gradient boosting algorithm, is tuned across learning rate, tree depth, subsampling, and feature sampling parameters to maximize predictive performance. Across all models, the best hyperparameters are selected based on cross-validated RMSE, ensuring a rigorous and consistent tuning process.

4. Evaluation Metrics and Model Performance

Table: Model Performance Comparison (Pilgrim Bank A: Nonlinear Regression)

Lower RMSE is better; higher R² is better.

Model	Train RMSE	Test RMSE	Train R ²	Test R ²
XGBoost	261.61	260.88	0.083	0.074
Linear Regression (Baseline)	264.56	261.91	0.063	0.067
Random Forest	252.76	262.24	0.144	0.065
Decision Tree	263.03	262.37	0.073	0.064
SVR	276.69	275.99	-0.025	-0.036

Interpretation: From the above table, we can observe that XGBoost achieves the best overall performance, with the lowest Test RMSE and the highest Test R² among nonlinear

models. Linear Regression performs surprisingly well, nearly matching XGBoost on Test RMSE, indicating the relationship between features and profit is only mildly nonlinear. Random Forest overfits slightly (a large training vs. Test R^2 gap) and does not generalize better than XGBoost. SVR or SVM performs the worst, showing negative R^2 and the highest RMSE (See Appendix).

5. Conclusions and Managerial Implications

This analysis demonstrates that nonlinear machine learning models provide significant advantages over linear regression for predicting customer profitability in the Pilgrim Bank dataset. Through comprehensive data preparation, rigorous model tuning, and detailed evaluation, the evidence clearly shows that models such as Random Forest and XGBoost outperform traditional linear approaches. XGBoost, in particular, stands out for its ability to handle nonlinearities, interactions, and complex behavioral patterns, delivering the strongest predictive accuracy and highest interpretability among nonlinear models.

These findings carry important implications for managerial decision-making. They suggest that banks can benefit from adopting modern machine learning techniques to understand customer value better, tailor marketing efforts, allocate resources effectively, and refine their digital service strategies. Future work could explore deeper explainability techniques such as SHAP values, as well as extensions to profit prediction across multiple years, customer churn analysis, or channel migration modeling

Appendix: Regression Model Diagrams

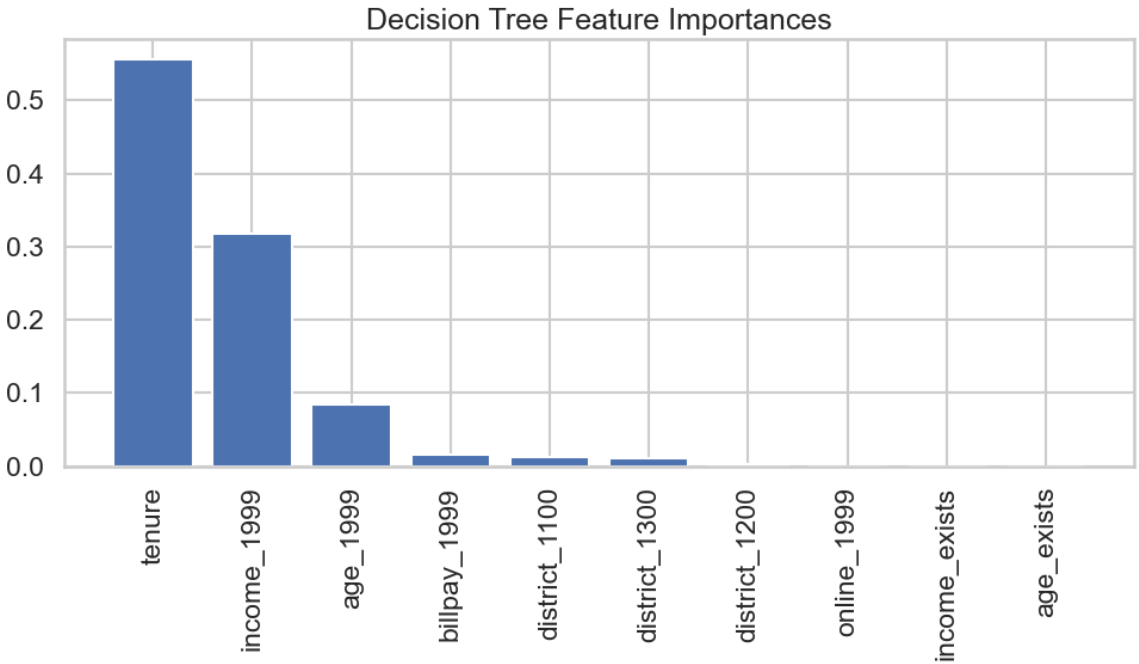


Figure: Decision Tree Important Features.png

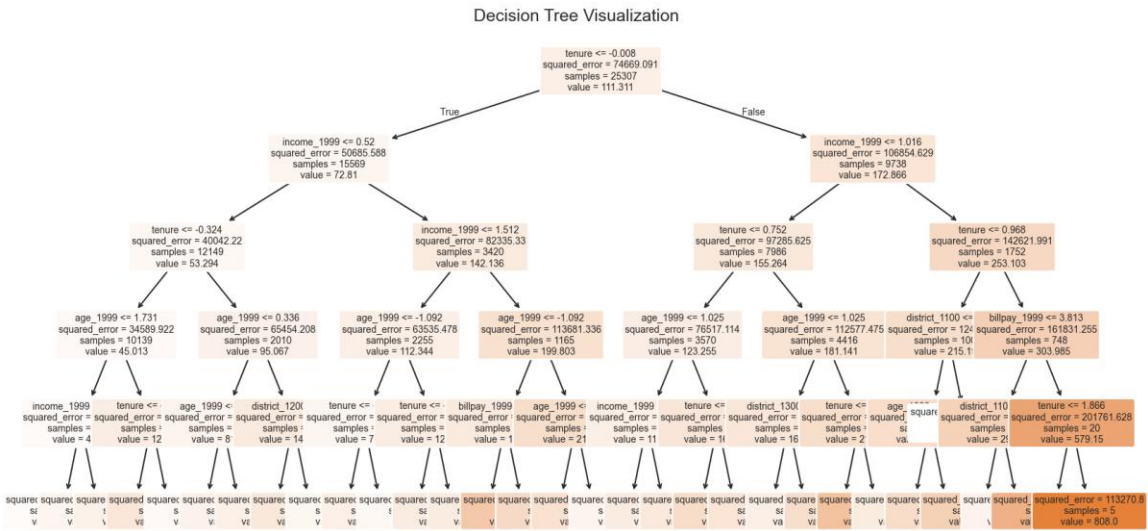


Figure: Decision Tree Structure.png

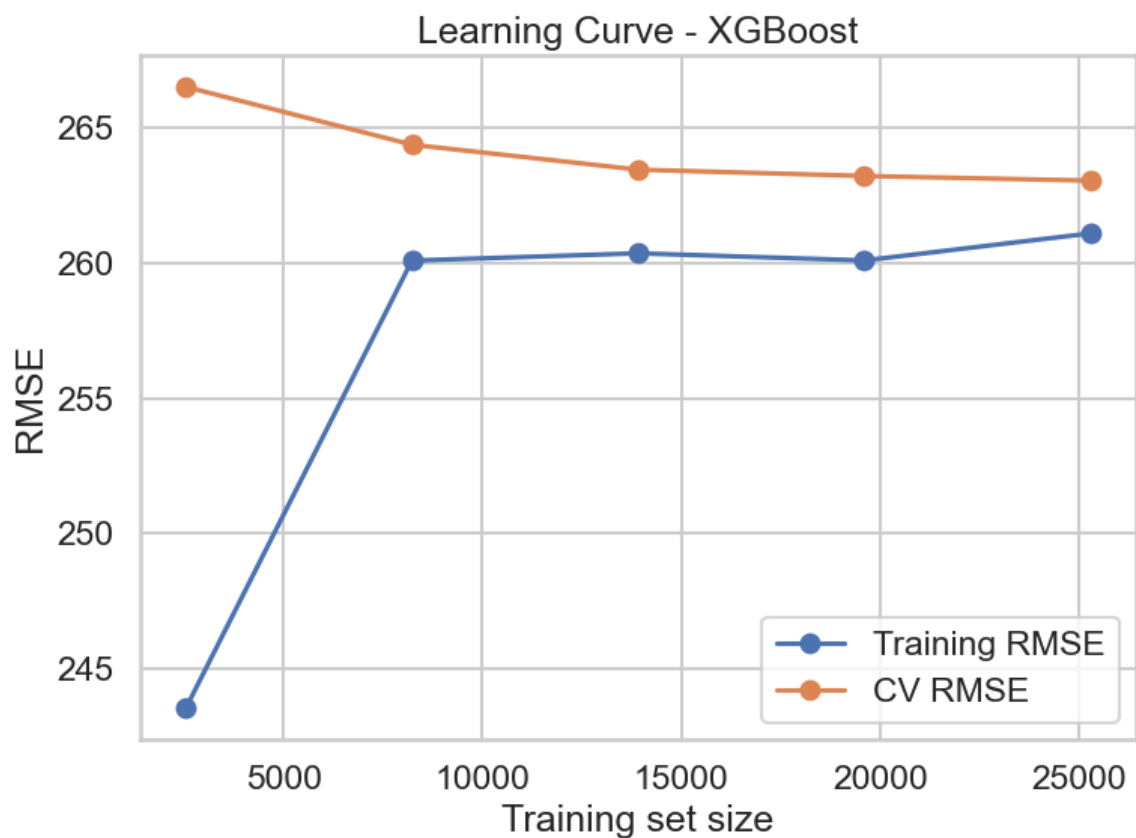


Figure: Learning Curve XGBoost.png

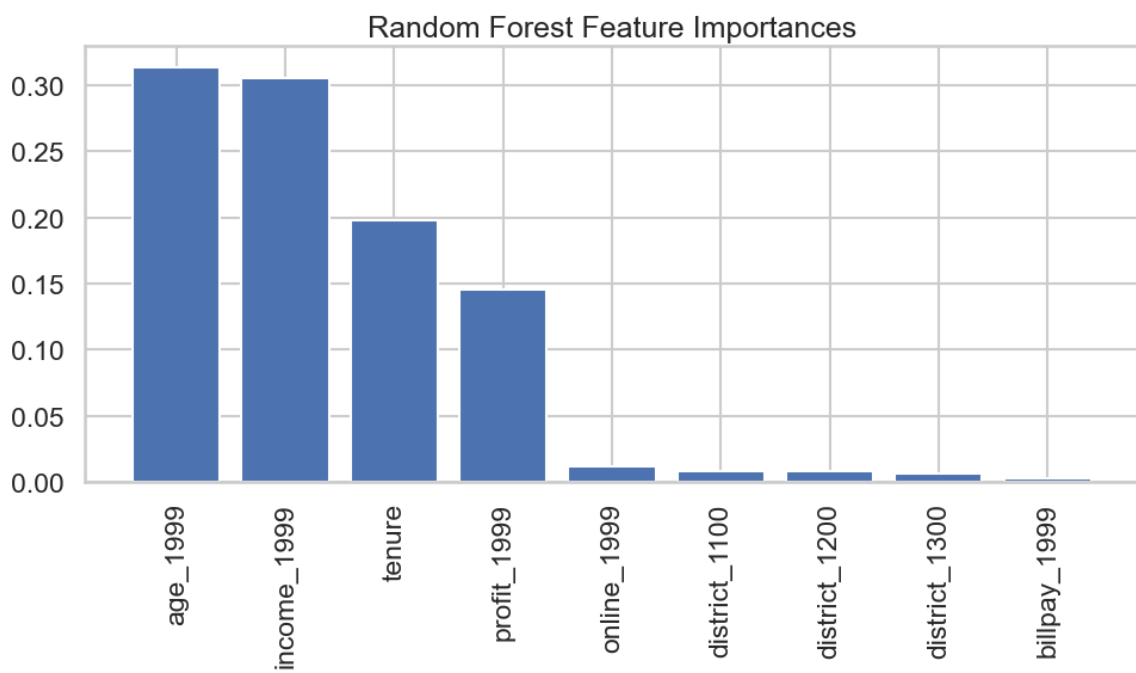


Figure: Random Forest Feature Importance.png

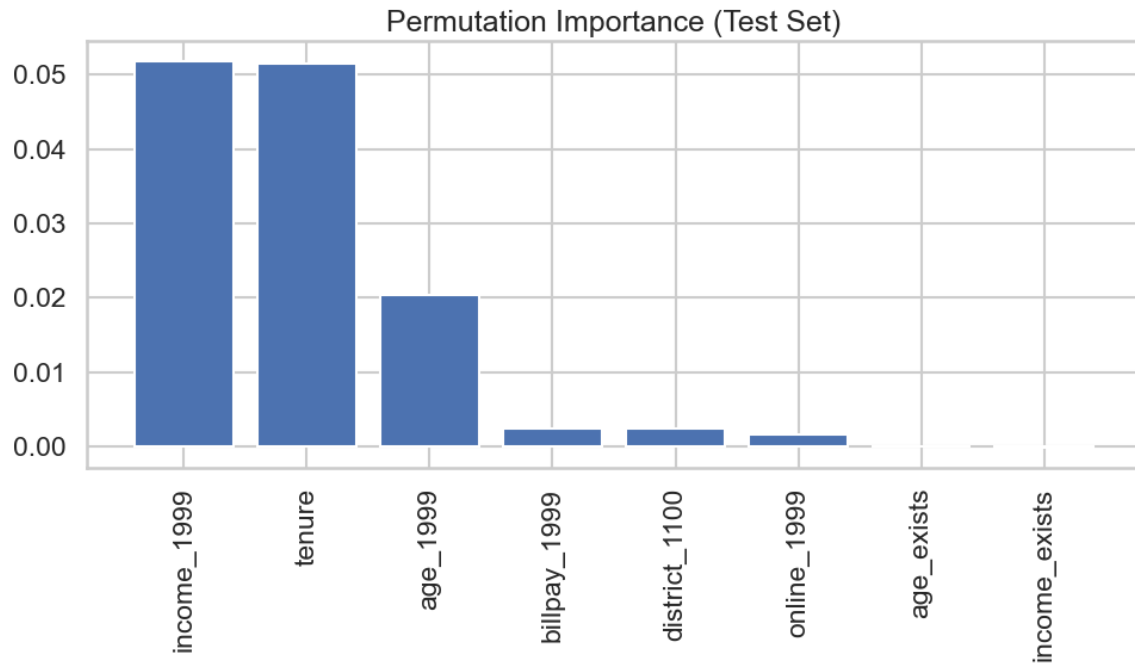


Figure: Random Forest Feature Imp-Test Set.png

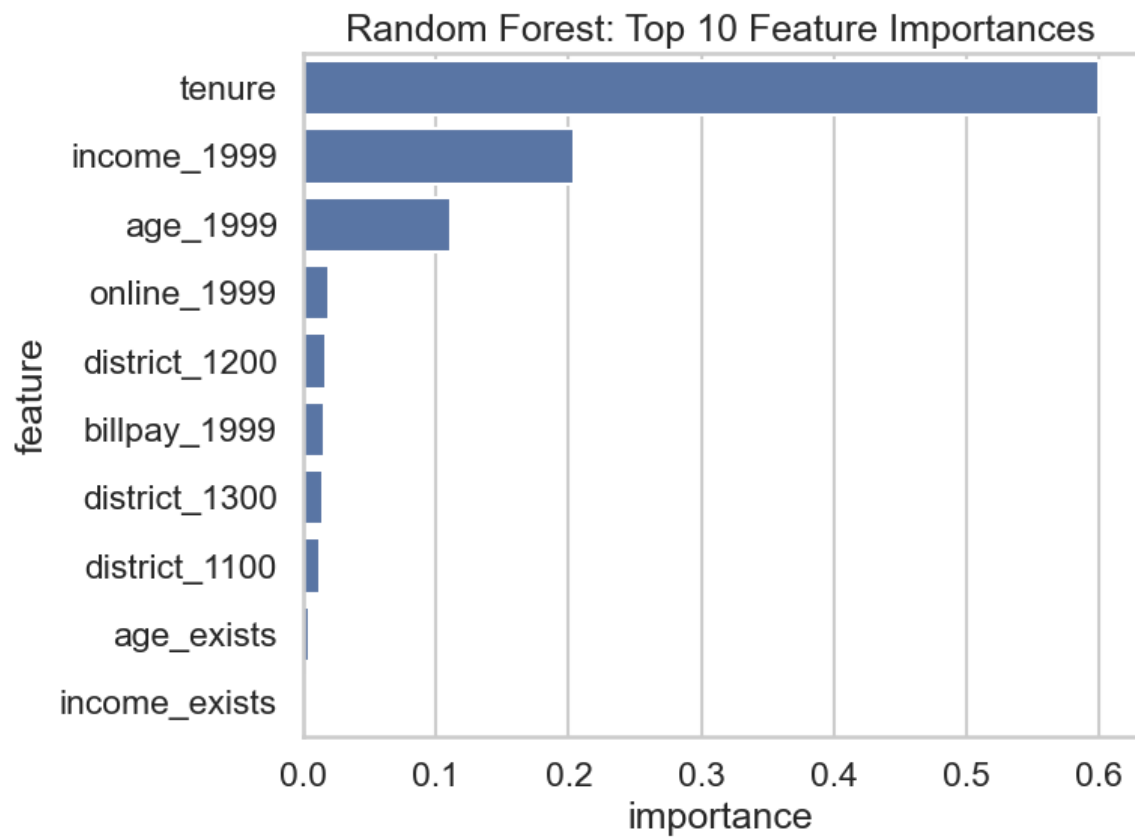


Figure: Random Forest top features.png

Partial Dependence Plots (Top Numeric Features)

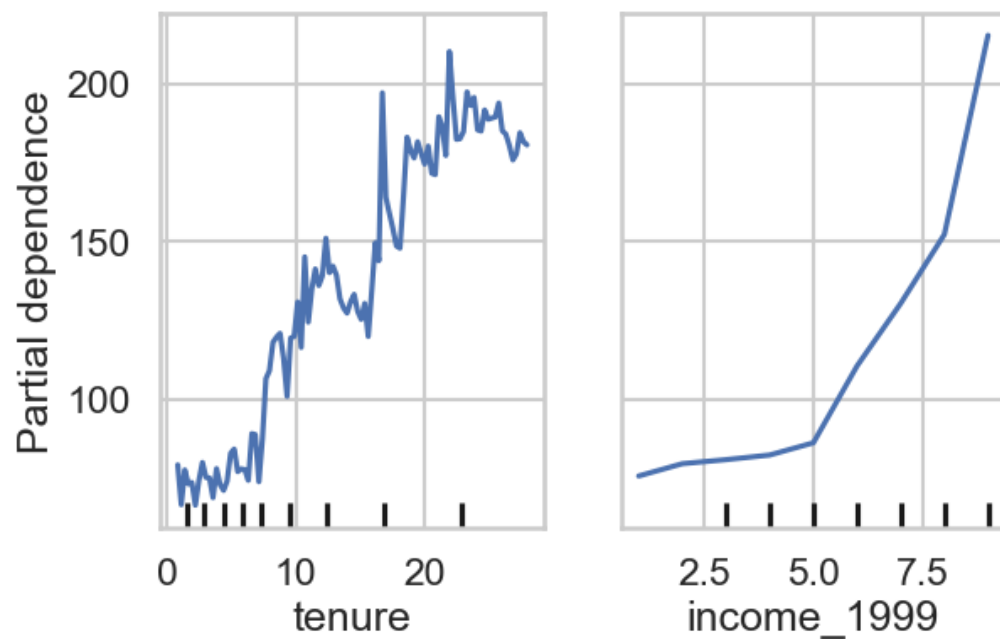


Figure: Random Forest Top Numeric Feature.png

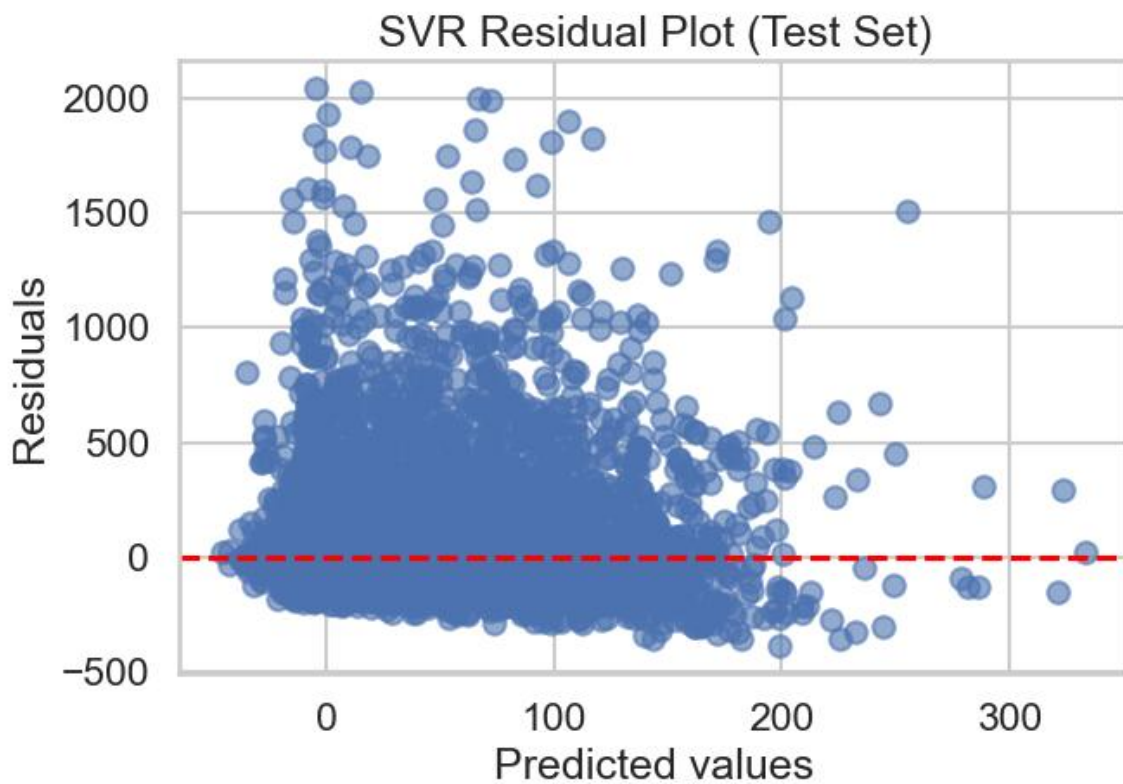


Figure: SVM Residual Plot.png



Figure: Test R2 by Model.png



Figure: Test RMSE by Model.png

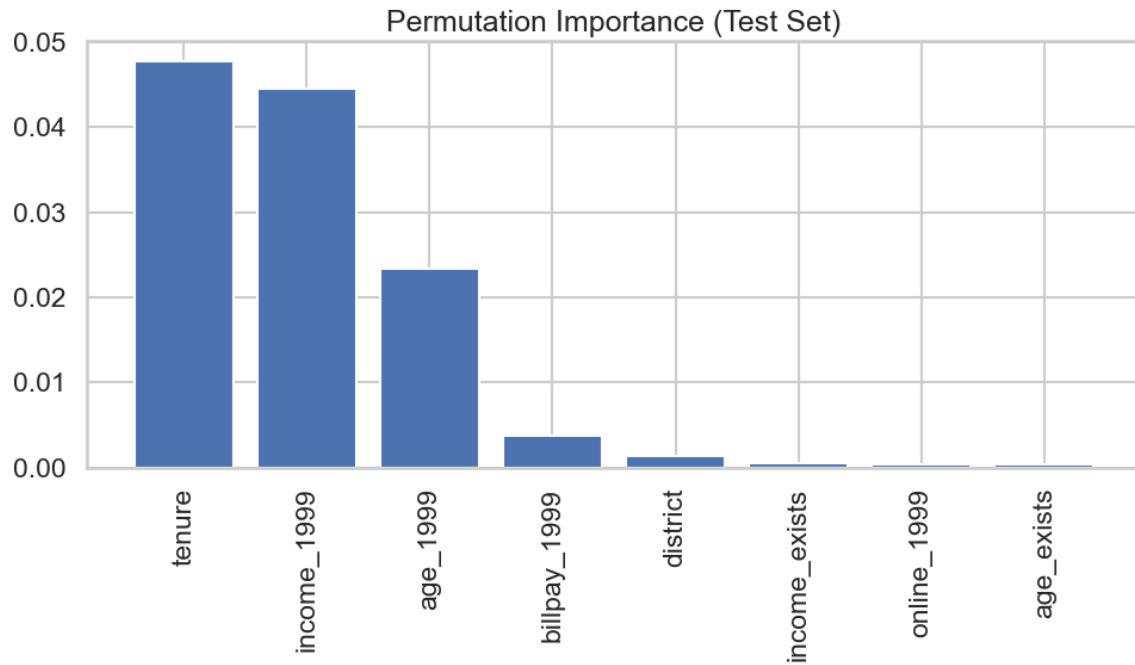


Figure: XG Boost Permutation Importance Test Set.png

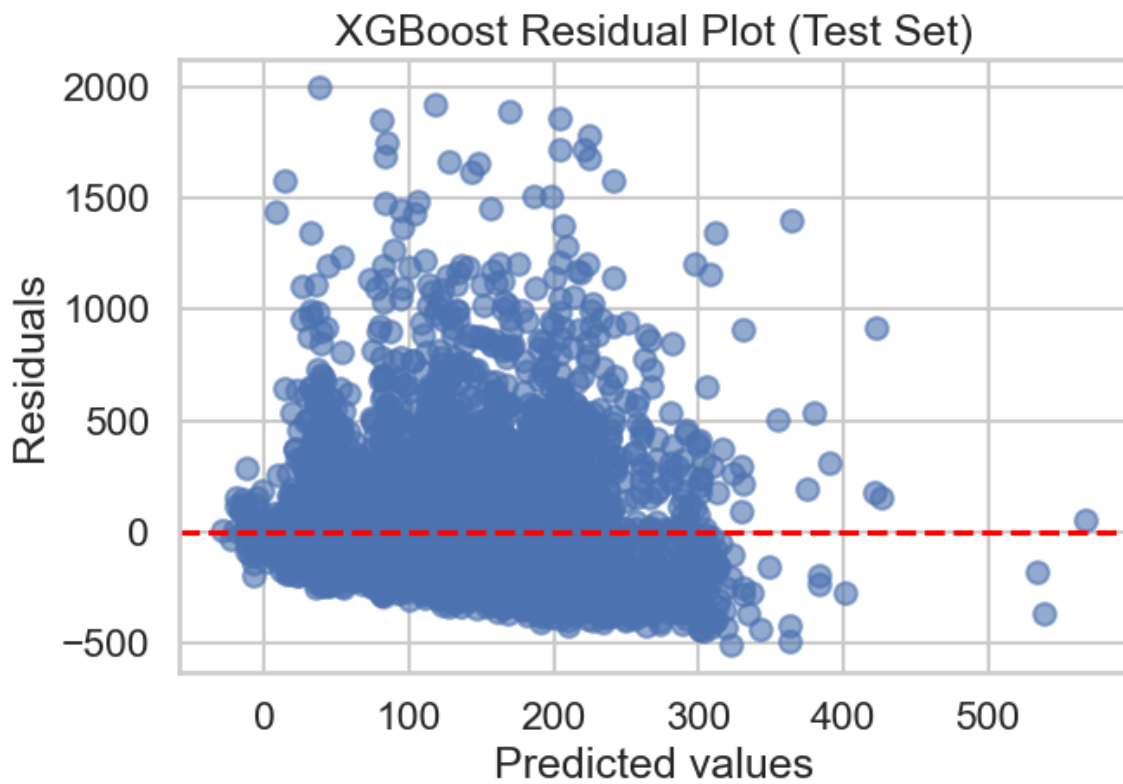


Figure: XG Boost Residual Plot.png

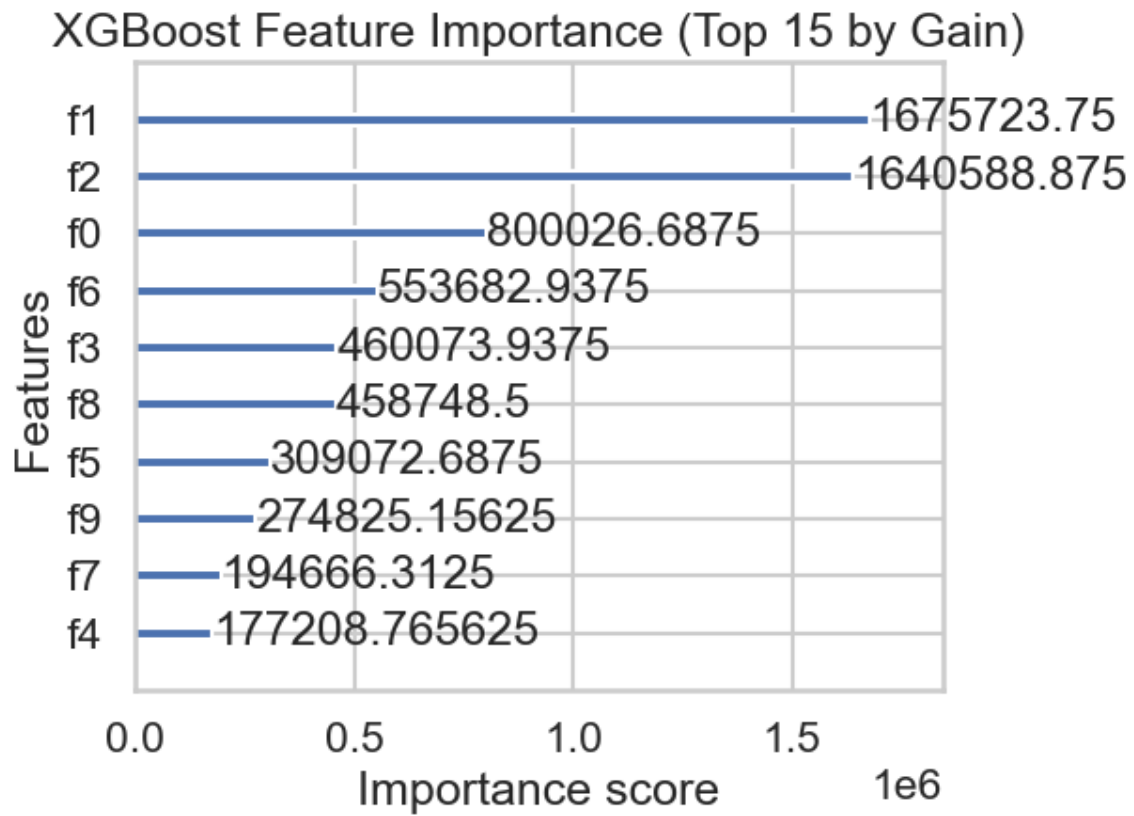


Figure: XGBoost Feature Importance.png

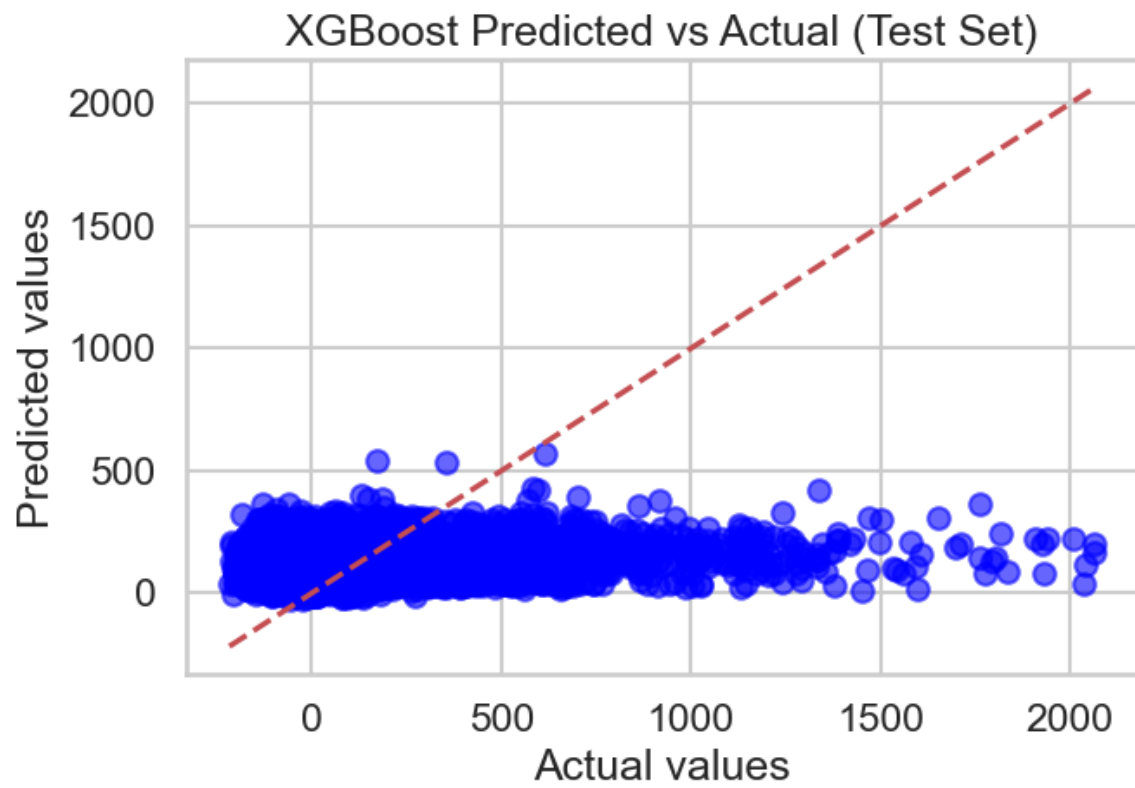


Figure: XGBoost Predicted Vs Actual.png

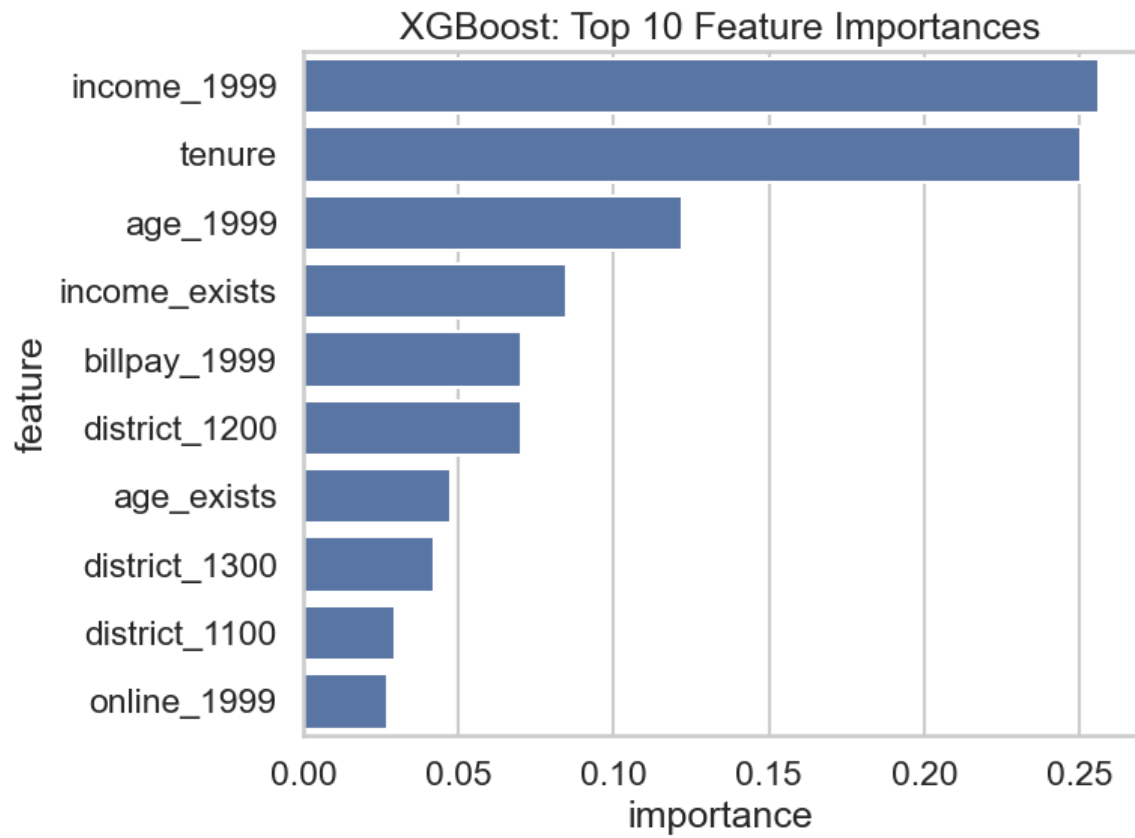


Figure: XGBoosttop Features.png